

Exchange rate and Production network

Brown bag seminar

Sugarkhuu Radnaa

Bonn Graduate School of Economics

June, 2026

Building on: Rubbo (2024) *Networks, Phillips Curves, and Monetary Policy*

Rubbo (2024): Key Results

Full model system (N sectors, Calvo $\hat{\delta}_i$, IO network Ω):

$$\text{IS: } \tilde{y}_t = \mathbb{E}\tilde{y}_{t+1} - \frac{1}{\gamma}(i_t - \mathbb{E}\pi_{t+1}^C - r_t^{\text{nat}})$$

$$\text{MC: } \mathbf{mc}_t = \alpha \mathbf{w}_t + \Omega \mathbf{p}_t - \log \mathbf{A}_t$$

$$\text{NKPC: } \boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t$$

$$\text{CPI: } \pi_t^C = \beta^T \boldsymbol{\pi}_t$$

$$\text{Policy: } i_t = r_t^{\text{nat}} + \phi_\pi \pi_t^C + \phi_y \tilde{y}_t$$

$$\mathcal{B} = \frac{(\gamma + \varphi)\Delta(I - \Omega\Delta)^{-1}\alpha}{\kappa}; \quad \boldsymbol{\chi}_t \equiv \log \mathbf{A}_t \text{ (TFP); } \mathcal{V} = \Delta(I - \Omega\Delta)^{-1}$$

Divine Coincidence (Prop. 1): unique index with *no* cost-push:

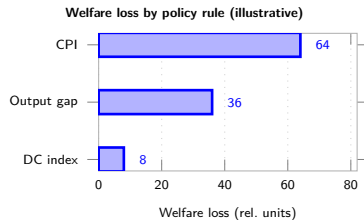
$$\pi_t^{DC} = \rho \mathbb{E}\pi_{t+1}^{DC} + \frac{\gamma + \varphi}{\Lambda^{DC}} \tilde{y}_t, \quad w_i^{DC} \propto \lambda_i \frac{1 - \hat{\delta}_i}{\hat{\delta}_i}$$

Unique: $\mathcal{V}$ rank $N-1 \Rightarrow$ one-dim. null space. Optimal policy = target π_t^{DC} .

Welfare (Prop. 3):

$$\mathbb{W} = \frac{1}{2} \sum_t \rho^t [(i) \tilde{y}_t^2 + (ii) \text{ within-sector} + (iii) \text{ cross-sector}].$$

DC targeting minimizes all three terms simultaneously.



DC targeting achieves the lowest welfare loss by eliminating cost-push distortion. Rubbo (2024) quantifies using US BEA IO data.

SOE Model: Structural Equations

Household (γ : risk aversion, φ : inv. labor elasticity)

$$\max \mathbb{E}_0 \sum_t \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right]$$

$$\text{s.t. } P_t^C C_t + B_t + E_t B_t^* = (1+i_{t-1})B_{t-1} + E_t(1+i_t^*)B_{t-1}^* + W_t L_t + \Pi_t$$

$$\text{FOC } B: \lambda_t = \beta(1+i_t) \mathbb{E}_t[\lambda_{t+1}/\pi_{t+1}^C]$$

$$\text{FOC } B^*: \lambda_t = \beta(1+i_t^*) \mathbb{E}_t[\lambda_{t+1}(E_{t+1}/E_t)/\pi_{t+1}^C]$$

$$\text{Labor: } W_t/P_t^C = C_t^\gamma L_t^\varphi$$

$\lambda_t \equiv C_t^{-\gamma}$; $\pi_{t+1}^C \equiv P_{t+1}^C/P_t^C$; $\text{FOC } B^* \div \text{FOC } B \Rightarrow \text{UIP below}$
International block (E_t : nominal ER, P_t^* : foreign price)

$$\text{LOP: } P_{Ft} = E_t P_t^*$$

$$\text{CPI: } P_t^C = \prod_i P_{it}^{\beta_i} \cdot (E_t P_t^*)^{\alpha_F}$$

$$\text{ToT: } \tau_t = E_t P_t^* / P_t^H$$

$$\text{UIP: } (1+i_t) = (1+i_t^*) \mathbb{E}_t[E_{t+1}/E_t] \cdot (1+\psi_t)/(1+\nu_t)$$

Firm i : Cobb-Douglas production, Calvo pricing

$$\text{Prod.: } Y_{it} = A_{it} L_{it}^{\alpha_i} \prod_j M_{ijt}^{\omega_{ij}} M_{Fit}^{\omega_{Fi}}$$

$$\text{MC: } MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\omega_{ij}} (E_t P_t^*)^{\omega_{Fi}}}{A_{it}}$$

$$\text{Calvo: } \begin{cases} \text{prob } (1-\hat{\delta}_i) : \text{ reset } \tilde{P}_{it} = \frac{\varepsilon}{\varepsilon-1} \mathbb{E}_t[\text{PV of MC}] \\ \text{prob } \hat{\delta}_i : \text{ keep } P_{i,t-1} \end{cases}$$

$$\text{Price: } P_{it}^{1-\varepsilon} = (1-\hat{\delta}_i) \tilde{P}_{it}^{1-\varepsilon} + \hat{\delta}_i P_{i,t-1}^{1-\varepsilon}$$

Market clearing

$$\text{Goods: } Y_{it} = C_{Hit} + X_{it} + \sum_j M_{jit}$$

$$\text{Labor: } L_t = \sum_i N_{it}$$

$\Rightarrow$ Log-linearize + take gaps around flex-price equilibrium
 $\Rightarrow$ IS, MC, NKPC (next slide).

Small Open Economy Extension

What I add: Import sector + UIP + law of one price. Exchange rate e_t enters marginal cost.

Full system (green = new SOE terms):

$$\text{IS: } \tilde{y}_t = \mathbb{E}\tilde{y}_{t+1} - \frac{1}{\gamma}(i_t - \mathbb{E}\pi_{t+1}^C - r_t^{\text{nat}}) + \Delta\tau_t$$

$$\text{MC: } mc_t = \alpha w_t + \Omega p_t + \omega_F(p_t^* + e_t) - \log \mathbf{A}_t$$

$$\text{NKPC: } \pi_t = \rho(I - \mathcal{V})\mathbb{E}\pi_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\chi_t + \Gamma\Delta e_t$$

$$\text{UIP: } i_t - i^* = \mathbb{E}\Delta e_{t+1} + \psi_t - \nu_t$$

$$\text{Policy: } i_t = r_t^{\text{nat}} + \phi_\pi \pi_t^C + \phi_y \tilde{y}_t; \nu_t^* : \text{FX intervention}$$

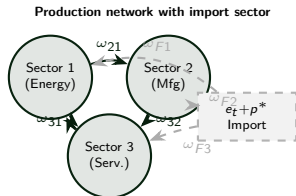
p_t^* : foreign price; i^* : foreign rate; e_t : log ER; ν_t : FXint

$\Gamma = \tilde{A}\omega_F/\kappa$, $\tilde{A} = \Delta(I - \Omega\Delta)^{-1}$, ω_F : import intensity shares

Changes from Rubbo:

- 1 IS: terms-of-trade effect $\Delta\tau = p^* + e - p^H$
- 2 MC: imported input costs enter price-setting
- 3 NKPC: ER cost-push $\Gamma\Delta e_t$ (network amplified)
- 4 UIP + ν_t : ER dynamics close the model

Two cost-push channels (TFP + ER) $\Rightarrow$ DC breaks down



Dashed: new SOE import flows. Depreciation ($\Delta e_t > 0$) raises imported input costs and propagates through the IO network via Γ .

Network Amplification: Upstream, Downstream, Centrality

Leontief expansion of ER pass-through Γ_i :

$$\Gamma_i = \frac{\hat{\delta}_i}{\kappa} \left[\underbrace{\omega_{Fi}}_{\text{direct}} + \underbrace{\sum_j \omega_{ij} \hat{\delta}_j \omega_{Fj}}_{\text{suppliers}} + \underbrace{\sum_{j,k} \omega_{ij} \hat{\delta}_j \omega_{jk} \hat{\delta}_k \omega_{Fk}}_{\text{2nd order}} + \dots \right]$$

Each round discounted by rigidities $\hat{\delta}$ along the chain.

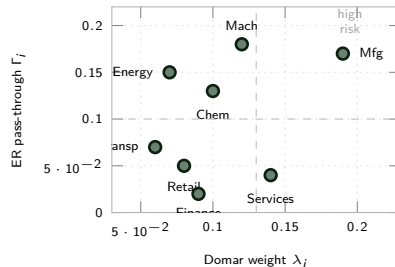
Upstream u : high $\omega_{Fu} \Rightarrow$ raises Γ_i for all customers; stickier upstream absorbs into markup, lower pass-through.

Downstream d : high Γ_d even if own ω_{Fd} low.

DC breakdown: requires $\omega_F \propto (I - \Omega \Delta) \phi^{DC}$ — fails empirically.

Welfare: $\lambda^T \Gamma = \sum_i \lambda_i \Gamma_i$ — Domar $\times \Gamma_i$ drives macro cost.

Centrality λ_i vs ER sensitivity Γ_i (illustrative)



Upper-right: high centrality + high import intensity — sectors where ER shocks generate the largest welfare costs.

Core Result: DC Breakdown and Managed Float

DC breaks down under float (Prop. 1'):

Cost-push-free index requires two conditions:

$$\underbrace{\phi^T \mathcal{V} = \mathbf{0}^T}_{\text{pins } \phi = \phi^{DC}} \quad \text{AND} \quad \underbrace{\phi^T \Gamma = 0}_{\text{generically fails}}$$

Two conditions, one unknown $\phi \Rightarrow$ no solution.

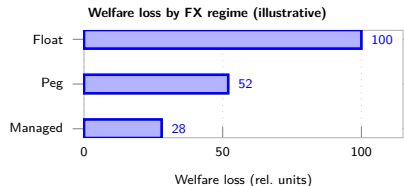
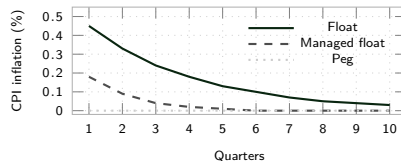
	Float	Peg	Managed
ER cost-push	Free	Zero	$\perp \phi^{DC}$
MP independence	Full	None	Full
DC survives?	No	Yes	Restored
Optimal	None	DC	DC + ν_t^*

Managed float: $i_t - i^* = \mathbb{E} \Delta e_{t+1} + \psi_t - \nu_t$

Optimal ν_t^* : $(\phi^{DC})^T \Gamma \Delta e_t(\nu_t^*) = 0$. Restores:

$\pi_t^{DC} = \rho \mathbb{E} \pi_{t+1}^{DC} + \frac{\gamma + \varphi}{\lambda^{DC}} \tilde{y}_t$. Two instruments, two targets.

IRF: CPI inflation response to 1% ER depreciation (illustrative)



Four Quantitative Experiments

1. **IRF to ER shock** — Simulate a 1% depreciation. Track sector-level CPI responses; plot λ_i vs Γ_i to identify which sectors drive aggregate inflation via the IO network.
2. **Welfare: float vs peg vs managed float** — Compare $\mathbb{W} = \frac{1}{2} \sum_t \rho^t [\tilde{y}_t^2 + \text{dispersion} + \text{misalloc.}]$ across three FX regimes for identical shocks. Quantifies the cost of DC breakdown under float and the gain from ν_t .
3. **DC breakdown test** — Measure the two cost-push channels:

$$\underbrace{\nu \chi_t}_{\text{TFP channel}} \quad \text{vs} \quad \underbrace{\Gamma \Delta e_t}_{\text{ER channel}}$$

If comparable in magnitude but not collinear, no single index eliminates both. Quantifies *how much* DC breaks down in calibrated data.

4. **Optimal FX intervention ν_t^*** — Solve for ν_t^* satisfying $(\phi^{DC})^T \Gamma \Delta e_t(\nu_t^*) = 0$, restoring the DC index under float. Two instruments (i_t, ν_t) , two targets (output gap, DC index).

Simulation Design and Roadmap

Code: extends Rubbo's Matlab directly

Module	Task
parameters_soe.m	Load import shares ω_F ; compute Γ
pol_fct_soe.m	Augment state with e_t , UIP
IRFs_soe.m	IRFs to ER & MP shocks
simul_soe.m	Monte Carlo: 3 FX regimes
welfare_soe.m	Add ER misallocation term

Augmented state space:

$$T_{\text{SOE}} = \begin{pmatrix} T_{\text{Rubbo}} & \Gamma \\ \mathbf{0} & \rho_e \end{pmatrix}, \quad e_t = \rho_e e_{t-1} + \varepsilon_t^e$$

Theory done:

- ✓ Rubbo lemmas/props verified
- ✓ SOE Phillips curve ($+\Gamma\Delta e_t$)
- ✓ DC breakdown (two conditions)
- ✓ Upstream/downstream decomp.
- ✓ Managed float (ν_t^*)
- ✓ Welfare SOE extension

Remaining (July 22):

- ☐ IO data: sector import shares ω_F
- ☐ Implement parameters_soe.m
- ☐ Augmented state space
- ☐ Welfare plots
- ☐ Paper sections 2–4

	Rubbo	This paper
Economy	Closed	SOE + ER
Cost-push	1 channel	2 channels
DC index	Unique	Breaks down
Optimal MP	DC target	DC + ν_t^*